

# Assessing Knit Parameters and Performance of Strain Sensors Made from Stainless Steel and Polypyrrole-based Yarns

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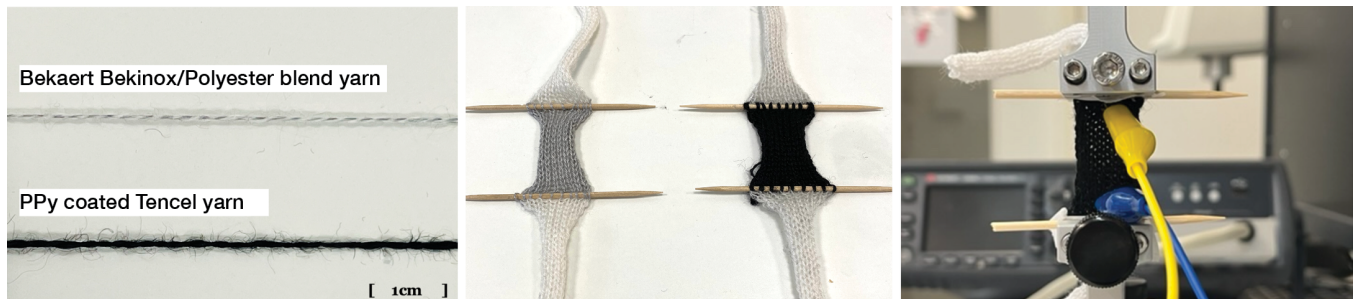
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**Figure 1:** (Left) The two conductive yarns used in the experiments. Bekaert Bekinox is a blend of 30% stainless steel and 70% polyester. The polypyrrole (PPy) coated Tencel yarn is a metal-free yarn which is lab-made. (Centre) Knitted sensor samples made from Stainless steel blend and PPy-coated yarn. (Right) Stretch test: PPy-based sample attached to a Stable Micro Systems TA.XTplusC texture analyzer.

## Abstract

Knitted strain sensors have emerged as a comfortable and user-preferred solution for wearable motion tracking. While knits made from silver-coated yarns are commonly used due to their high conductivity, their susceptibility to degradation in sweat and environmental concerns limit their practicality. This study investigates two alternatives: stainless steel-blend and metal-free polymer-coated yarns, specifically polypyrrole (PPy) on Tencel. We evaluate optimal knit parameters and compare the performance of sensors made from each material. Findings suggest that a dense stitch with at least 30 rows and wale-oriented stress improves sensor performance. The polypyrrole-coated yarn demonstrated superior signal stability and linearity, indicating its promise as an environmentally favourable and reliable material for future wearable strain sensors.

\*Led the study implementation.



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## CCS Concepts

• **Hardware** → **Sensors and actuators.**

## Keywords

e-textiles, soft sensors, knitting, stretch sensing, wearable, sensor characterisation

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## 1 Introduction

Using textile-based sensors for wearable motion tracking applications, such as following breathing patterns and motion of joints [15], have been shown to improve the comfort of use in longer sessions and the coverage area for sensing. Some results have even

indicated that users prefer smart textiles over more established tracking systems [3].

Knitted strain sensors made from conductive yarns is a popular choice for making textile-based motion trackers, however, no consensus has been established on the optimal material, method, form, or shape for applying these sensors. One of the most studied materials in the latest textile-based motion-tracking studies is silver-blended yarn due to its good conductivity [10, 11]. However, silver can be corroded by sodium chloride solutions, making sensors vulnerable to sweat, especially in sports applications [13]. In addition, the environmental impact and price of silver are unfavourable when considering the market value [16].

As an alternative, we present initial results from testing two non-silver conductive yarns as more environmentally favourable options for constructing knitted sensors. The first is stainless steel blend yarn, which is less conductive but offers a widely available, corrosion-resistant option with less environmental impact and a lower price [4]. And the second is a custom-made metal-free yarn, made by coating polypyrrole (PPy) onto biodegradable Tencel [6]. PPy, as a conductive polymer, presents a compelling alternative to metal-based yarns in wearable textiles due to its ease of processing, cost-effectiveness, robustness, biocompatibility, and comparatively lower environmental impact [1, 7, 14]. We conduct two studies: First, we construct a series of knitted sensor samples and evaluate different knit parameters to find the optimal settings for constructing knit sensors. Secondly we compare sensing performance of sensors knitted with stainless steel blend yarn and the PPy-coated yarn. Our results recommend a denser stitch, at least 30 rows, and wale-oriented stress for wearable sensors like joint monitors. Sensors made from PPy samples show greater stability and linearity than those made with steel yarn.

## 2 Study overview

### 2.1 Working principle of knitted strain sensors

Knitted strain sensors are resistive sensors made from knitting conductive yarns. When the knitted structure is stretched, the contact points between the loops of yarn get held tightly together, increasing the surface area of contact and thus decreasing its resistance. Increasing the rows of the knit increases resistance, while increasing columns decreases resistance [9].

### 2.2 Sample construction and evaluation

The samples were knitted using a Silver Reed SK840 knitting machine, which is a commercially available knitting machine operated manually to create two-dimensional knitted structures. The carriage knits the yarn through the needles, but the cast-on, cast-off, and carriage operations are done by hand. A standard weft knit is produced by moving the carriage manually from left to right.

A total of 11 samples were made to compare different knit parameters (stitch size, stretch direction, sensor size) and yarn materials. Stitch size of the knit refers to the size of the loops constituting the knit fabric and can be adjusted using the tension dial on the knitting machine. Stretch direction here refers to the two primary directions in which the knitted structure can be stretched- the wale direction (length-wise) and the course direction (width-wise). The sensor size

is determined by the number of rows and columns knitted in the sample.

### 2.3 Yarn materials

This study used two types of yarn (Figure 1-left). The first is stainless steel polyester blend by Bekaert consists of 70% polyester and 30% Bekinox stainless steel [2]. This is a commercially available yarn with resistance of 8  $\Omega$ /cm. The second is a conductive PPy-coated yarn, made by chemical polymerisation of pyrrole onto a base yarn made of Tencel [17]. This type of coating causes the conductive material to penetrate the cross-section of the yarn, leading to a minimum resistance of 22.3  $\Omega$  /cm [6], although not perfectly uniform due to the coating process. Furthermore, PPy does not undergo corrosion and is often used as a protective layer for metal components [8]. The PPy-coated yarn is not commercially available and has limited availability due to lab processes. Both yarns have similar yarn thickness; thus, knitted samples from each material having the same stitch count resulted in the same sample size.

PPy-based yarn, being limited in supply, was used only for material comparison. Bekaert yarn was used to test different knit parameters and sensors with optimal settings were later knitted with both steel and carbon yarn for performance comparison.

### 2.4 Testing method

**2.4.1 Circuit and voltage measurements.** Voltage measurements of the knitted sensors were performed using a voltage divider circuit with a 220  $\Omega$  reference resistor and an Arduino MKR WiFi 1010 board. The Arduino supplied 3.3 V via the VCC pin, and the circuit operated with a DC current of 7 mA. Voltage values were measured through an analog pin and recorded along with corresponding time indices to a comma-separated values (CSV) file using Python, with a sampling frequency of 200 points per second (pps).

**2.4.2 Sample stretching.** Cyclic stretches were performed using a TA.XTplusC texture analyzer with clamp probes (Figure 1-right) securing the knitted sample. A wooden knitting needle was inserted between the clamps to improve grip and prevent edge rolling. The setup was operated via the previously described Arduino system.

### 2.5 Sensor characterisation

Sensor characterisation assessed value range, stability, and sensitivity using gauge factor (GF), hysteresis area, and value drift. GF indicates sensitivity, with higher values reflecting greater responsiveness [5]. Hysteresis measures response delay and reflects repeatability [12]. Value drift refers to baseline shifts over time, signaling reduced accuracy.

## 3 Results: Comparison of knitting parameters

Stitch size 1 outperformed sizes 2 and 3 (Figure 2). Looser structures, especially size 3, were more prone to deformation and value drift under repeated stress. Size 1 showed the least change and best recovery, while size 3 exhibited significant deformation and near-zero sensitivity after 50 cycles. Its gauge factor dropped to zero, indicating a loss of sensor function. Stretching in the wale direction showed more stable behaviour (Figures 3 and 6). The course direction produced greater value changes and inconsistent

| Stitch Size   | Value drift (50 cycles, 80% stretch) |               |               | Gauge factor (50 cycles, 80% stretch) |               |               |
|---------------|--------------------------------------|---------------|---------------|---------------------------------------|---------------|---------------|
| Stretch speed | Stitch size 1                        | Stitch size 2 | Stitch size 3 | Stitch size 1                         | Stitch size 2 | Stitch size 3 |
| 5 mm/s        | 88% to 36%                           | 78% to 5%     | 53% to 1%     | 2.63 / 1.03                           | 2.21 / 0.04   | 1.45 / 0      |

Figure 2: Value drift and gauge factor observed in knitted samples (30 rows x 40 columns) of different stitch sizes.

| Stress Direction | Value drift (50 cycles, 80% stretch) |                  | Hysteresis area (50 cycles, 80% stretch) |                  |
|------------------|--------------------------------------|------------------|------------------------------------------|------------------|
| Stretch speed    | Wale direction                       | Course direction | Wale direction                           | Course direction |
| 5 mm/s           | 92% to 51%                           | 18% to 48%       | 204.560                                  | 984.311          |

Figure 3: Value drift and hysteresis area observed in knitted samples (30 rows x 40 columns) stretched in different directions.

| Sensor size   | Value drift of 40 columns (50 cycles, 80% stretch) |            |            |            | Value drift of 30 rows (20 cycles, 80% stretch) |            |            |            |
|---------------|----------------------------------------------------|------------|------------|------------|-------------------------------------------------|------------|------------|------------|
| Stretch speed | 80 rows                                            | 40 rows    | 30 rows    | 20 rows    | 20 columns                                      | 15 columns | 10 columns | 5 columns  |
| 5 mm/s        | 67% to 6%                                          | 84% to 24% | 92% to 51% | 75% to 32% | 51% to 14%                                      | 52% to 9%  | 48% to 15% | 66% to 28% |

Figure 4: Value drift observed in sensor samples with varying number of columns and rows of stitches.

| Material comparison | Value drift (50 cycles, 80% stretch) |                | Hysteresis area (50 cycles, 80% stretch) |                |
|---------------------|--------------------------------------|----------------|------------------------------------------|----------------|
| Stretch speed       | Stainless steel yarn                 | PPy-based yarn | Stainless steel yarn                     | PPy-based yarn |
| 5 mm/s              | 75% to 32%                           | 20% to 18%     | 359.369                                  | 136.016        |

Figure 5: Value drift and hysteresis area observed in knitted samples (20 rows x 10 columns) made from stainless steel and PPy-based yarns

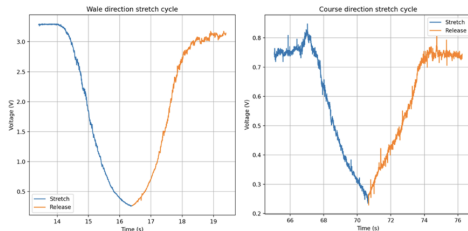


Figure 6: Hysteresis comparison of knitted sample (30 rows X 40 columns) showed the wale direction had a smoother, triangular cycle with lower hysteresis. The course direction showed more noise and flat segments, indicating higher hysteresis and reduced linearity.

results under identical conditions. However, it is important to note that in practice, sensors experience multi-directional stretching across 3D surfaces, not just in one direction.

Regarding sensor size, optimal performance was observed with sensors around 30 rows (Figure 4). Samples with 80 rows showed visible deformation, while 20-row sensors exhibited increased voltage noise. In contrast, reducing the number of columns to as few as five had little effect on noise levels. Among narrower designs, the five-column sensor provided the most stable sensitivity, though an optimal column count could not be identified. The number of columns could be chosen depending on placement; for example,

| Material comparison | Gauge factor (50 cycles, 80% stretch) |                |
|---------------------|---------------------------------------|----------------|
| Stretch speed       | Stainless steel yarn                  | PPy-based yarn |
| 5 mm/s              | 2.95 / 1.21                           | 1.23 / 1.13    |

Figure 7: Gauge factor observed in knitted samples (20 rows x 10 columns) made from stainless steel and PPy-based yarns

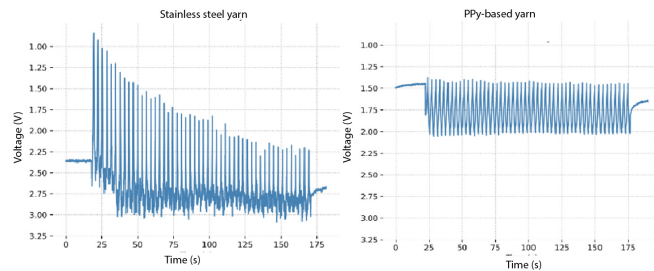
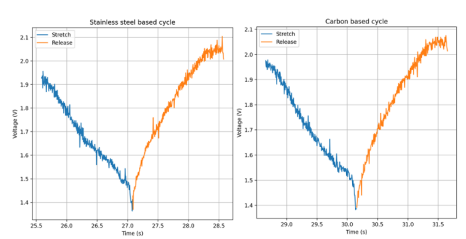


Figure 8: In a 50-cycle 80% stretch test, the steel sensor showed high baseline noise and significant drift, while the PPy sensor had minimal noise and maintained a consistent value range.

sensors positioned over the elbow should be wide enough to ensure proper alignment and secure placement over the joint.



**Figure 9: Stretch cycle comparison between Steel and PPy knit samples. Although both samples exhibit noise, the PPy-based sample has a more balanced stretch cycle loop, with more identical sides. The stainless steel blend cycles had more uneven cycles, resulting in the disproportionate shape of the release voltage increasing more quickly.**

#### 4 Results: Material comparison

Sensor knit parameters for the material comparison were based on findings from the previous study. The sensors used a plain weft of stitch size of 1 and wale-oriented stress direction (Figure 1-centre). However, due to the limited length of the PPy-based yarn, the sensor size of 20 rows by 10 columns was used.

The PPy-based sensor showed less deformation and value drift over time compared to the stainless steel sensor, which exhibited declining values (Figure 5). It also demonstrated better linearity with a smaller hysteresis area (Figure 9). Although the stainless steel sensor had a higher gauge factor, its performance degraded more quickly due to deformation (Figure 7). Additionally, the carbon sensor was more stable, with minimal noise, whereas the steel sensor displayed significant signal noise (Figure 8).

#### 5 Conclusion

Our initial tests demonstrate that while stainless steel yarn-based sensors can detect incremental motion and stretching to some degree, issues with value drift and structural deformation limit their suitability for applications involving frequent or intense movement. In contrast, PPy-coated yarn offers a more promising alternative for knitted strain sensors due to its greater resistance to deformation, improved value stability, and more linear response.

Although care was taken to make all the samples as uniformly as possible, some variations could have occurred during the knitting process with the manual knitting machine. Looking ahead, we aim to scale up the production of PPy-based yarns and further standardize the fabrication of the knitted fabrics. This will enable more comprehensive testing in real-world garment applications, allowing us to better evaluate their practical effectiveness.

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